Project Design Phase-I
Solution Architecture

Date	05 March 2026
Team ID	SWUID202642221
Project Name	Gruha Alankara: Interior Design Platform with AI Design Agent and Augmented Reality Visualization
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a structured process that connects **business needs with appropriate technology solutions**. It defines how different components of a system interact to deliver the required functionality. For the **Gruha Alankara – AI Interior Design Platform with AR Visualization**, the solution architecture outlines the overall system structure, technology stack, and data flow between the user interface, backend services, AI models, and database systems.

The architecture ensures that the platform can efficiently process room images, generate intelligent interior design suggestions, and allow users to visualize furniture placement using augmented reality. It integrates modern technologies such as **AI models, computer vision, AR visualization, and web frameworks** to provide a seamless and interactive user experience.

The main goals of the solution architecture for the Gruha Alankara system are:

• **Identify the best technological solution** to address the challenge of providing accessible and intelligent interior design recommendations without requiring professional designers.

• **Describe the overall system structure**, including the interaction between the frontend application, backend server, AI design engine, AR visualization module, and the database used for storing user data and design outputs.

• **Define system features and development phases**, such as user authentication, room image upload, AI-based room analysis, theme recommendations, augmented reality furniture placement, and design storage for future reference.

• **Provide clear technical specifications** that guide how the system should be designed, implemented, tested, and maintained, ensuring that all components work together effectively to deliver personalized interior design solutions.

In the Gruha Alankara platform, users interact with a **web-based interface** where they can upload room images or capture them using a camera. These images are processed by the **backend server**, which communicates with the **AI design engine** to analyze room features and generate interior design suggestions. The results are stored in a **SQLite database** and returned to the user interface, where they can be visualized using **AR technology** to simulate furniture placement within the user's room environment.

This architecture enables the system to deliver **intelligent design recommendations, immersive visualization, and efficient data management**, ultimately making interior design more accessible, interactive, and user-friendly.

Solution Architecture Diagram:

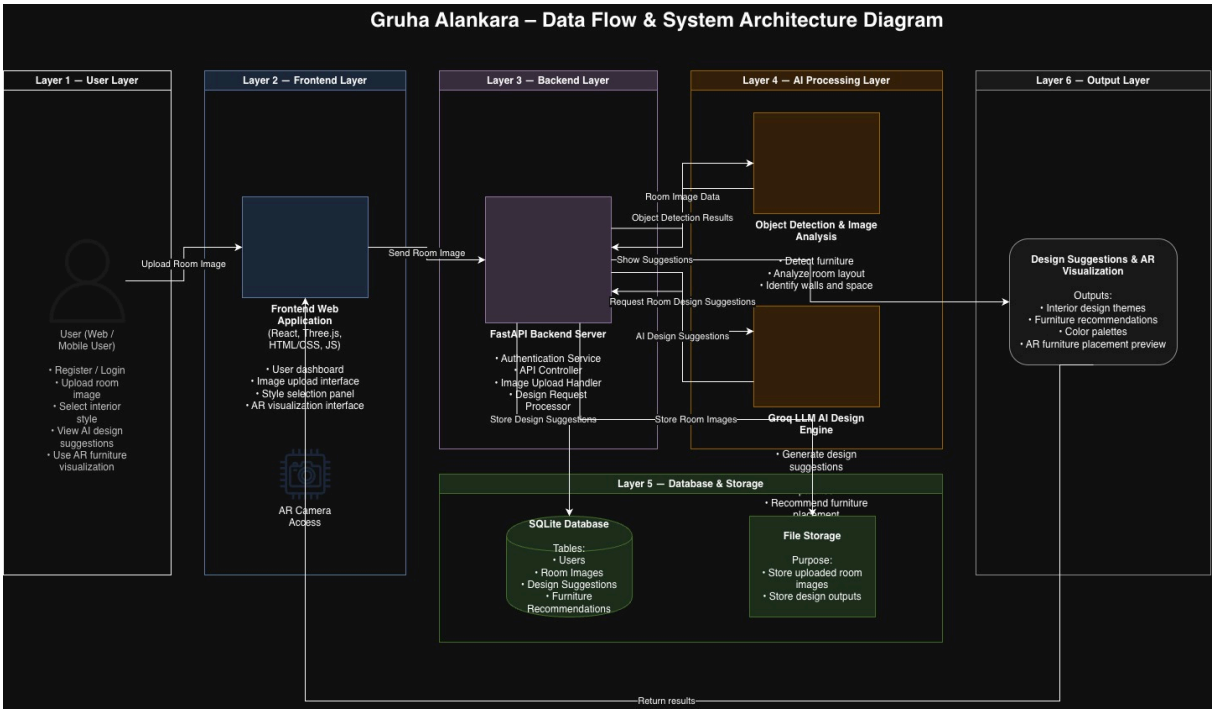

Figure 1: Architecture and data flow of the Gruha Alankara: Interior Design Platform with AI Design Agent and Augmented Reality Visualization